

More AI Reviewers Help — *Only If Their Mistakes Differ*

Companion to: “Evaluating Decorrelation in LLM Code Review: A Pre-Registered Study”

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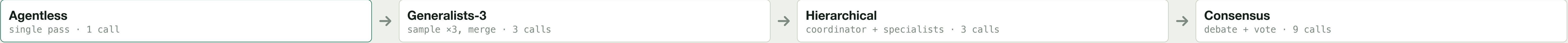
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01 MOTIVATION

LLMs increasingly automate code review, and frameworks split the work across specialized agents — managers, specialists, voters — and price accordingly.

Does the *structure* of that collaboration improve review quality, or just cost more? We hold the model, PR snapshot, prompts, and pipeline fixed and vary only structure. Pre-registered on OSF; prompts frozen before any confirmatory data.

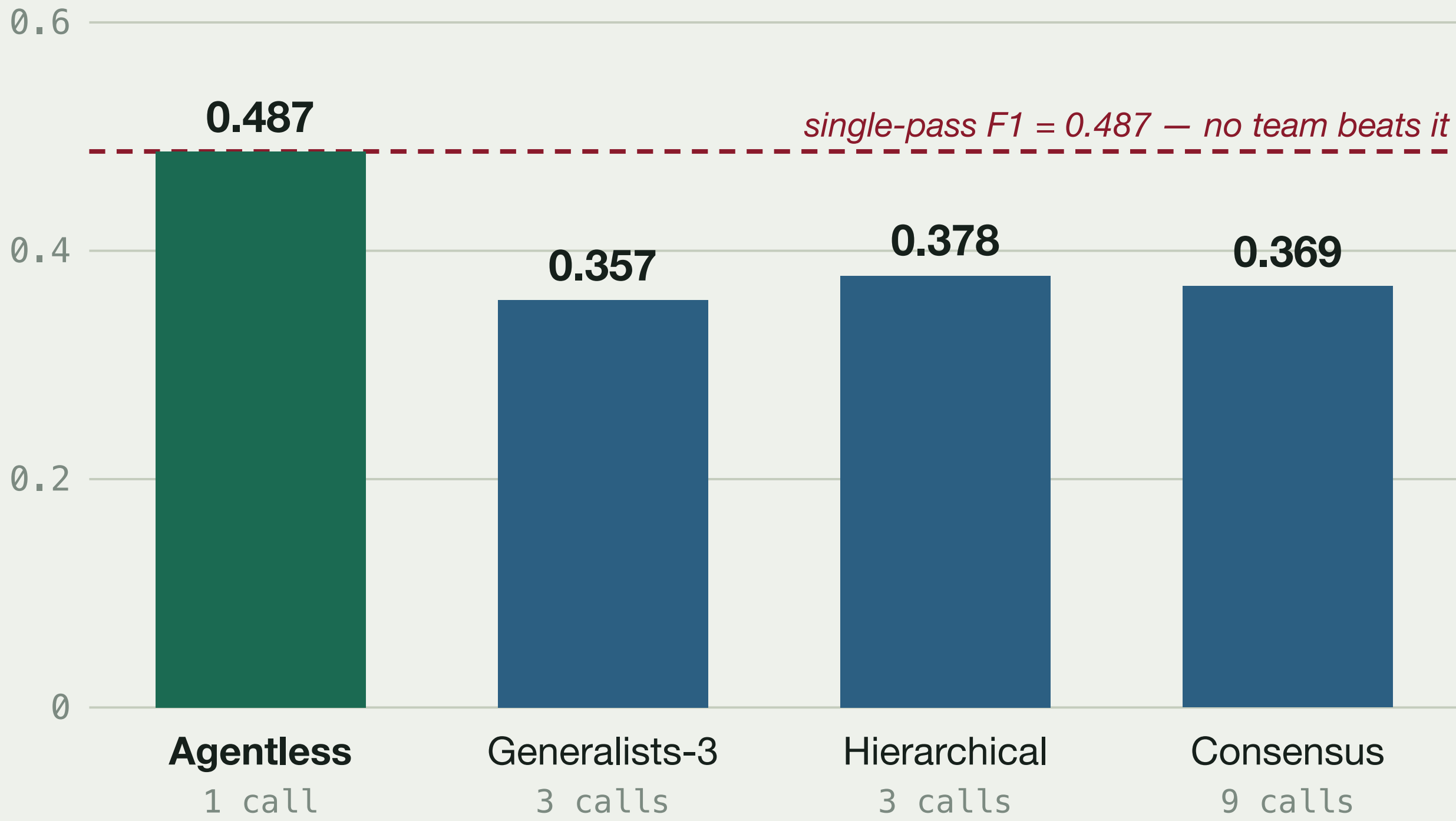
02 METHOD — A FOUR-RUNG LADDER, EACH RUNG ADDS ONE CAPABILITY



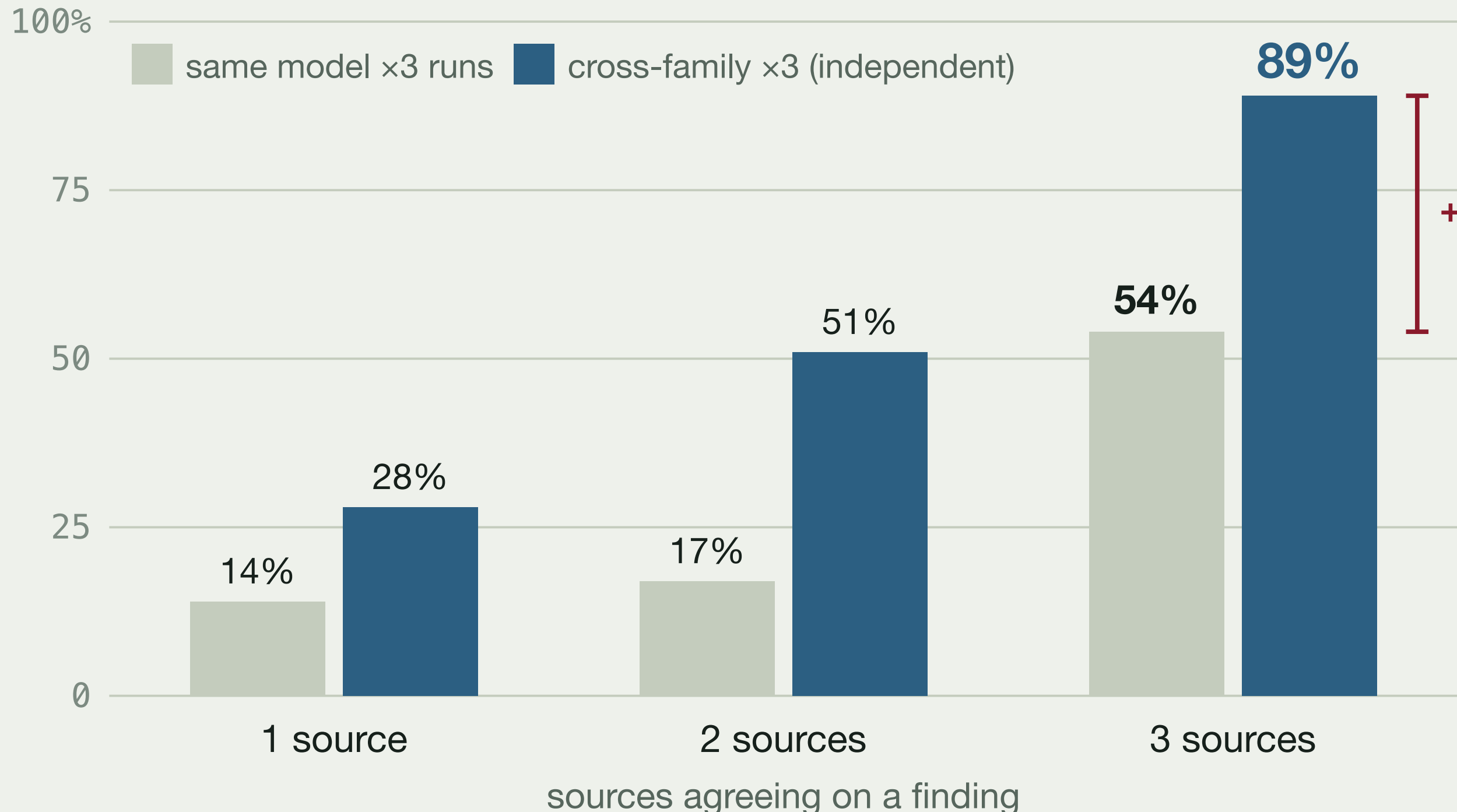
Adjacent contrasts isolate **sampling** vs **specialization** vs **communication**. Model held constant: Claude Haiku 4.5. Evaluated on three datasets:

E1 · Qodo 99 PRs · 1,188 reviews · injected ground truth	E2 · SWE-PRBench 50 PRs · 600 reviews · human-reviewer agreement	E3 · Industrial 30 PRs · 627 reviews · no oracle
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03 TWO RESULTS: MORE AGENTS DON'T HELP — DECORRELATION DOES



The null (myth-buster). No multi-agent arm beats single-pass F1. Extra agents raise recall (0.62–0.63 vs 0.50) but halve precision — buying recall at 3–9× the calls and 5× the cost per confirmed finding.



The signal (decorrelation). Three *independent* model families match injected ground truth 89% of the time; one model repeated three times saturates at 54%. Independence — not more compute — is what makes agreement predict truth.

04 KEY FINDINGS (PRE-REGISTERED, CONFIRMATORY)

- Specialization is null:** Hierarchical ties Generalists-3 on recall (0.624 vs 0.626, $p=0.73$).
- No arm beats single-pass F1** (0.487 vs 0.357–0.378; Holm $p<0.001$).
- Communication unsupported:** Consensus does not beat Hierarchical (Holm $p=0.12$) at 5× the cost.
- Grounding bounds recall at a ≈0.83 ceiling** — residue is linter-recoverable conventions + an execution-reachable functional core.
- Self-consistency fails; cross-family agreement lifts precision.**

06 WHY IT MATTERS

- Don’t deploy multi-agent review expecting a quality lift — you buy recall, not F1, at 3–9× the cost.
- Decorrelate verifiers: independent model families, not one model repeating itself.
- Measure benchmark completeness — a 0.83 recall ceiling means much “imprecision” is the benchmark’s.
- Treat LLM-judged evaluation on real code as unreliable ($\kappa = 0.03$).

07 NEXT STEPS · OPEN SCIENCE

- Tool-grounded verifiers — execution + static analysis as a correctness oracle.
- All runs, ablations, and artifacts released for replay.

05 WHERE RESEARCH MEETS THE CLIENT: E3 IS THE INDIGENOMICS RAP PLATFORM

Our E3 industrial dataset *is* the live RAP data platform we built for the **Indigenomics Institute** — its **30 real merged pull requests, 627 reviews**. On this unlabeled industrial code the verification relationship reaches its *ecological boundary*: LLM correctness judges themselves disagree (**$\kappa = 0.03$**), so the decorrelation signal that holds on labeled benchmarks cannot be validated without an oracle. The research and the client’s product are one system.